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$$F(x,y) = x^2 + 3xy + y^2$$

1. $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$ compute first order partial derivatives

$\therefore \frac{\partial F}{\partial x} = 2x + 3y$

$\therefore \frac{\partial F}{\partial y} = 3x + 2y$

2. $\nabla F(x,y) = \left[\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}\right]$

$= [2x + 3y, 3x + 2y]$

3. at $(2,1)$

$\nabla F(2,1) = [2(2)+3(1), 3(2)+2(1)]$

$= [7,8]$

4. directional ~~direct~~ derivative in direction of $v = [\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

dot product,

$D_v F = \nabla F \cdot v$

$= [7,8] \cdot \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

$= \frac{7}{\sqrt{2}} + \frac{8}{\sqrt{2}}$

$= \frac{15}{\sqrt{2}}$

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5. $\dfrac{\partial F}{\partial r}$

$x = r\cos\theta$
$y = r\sin\theta$

$$\frac{\partial F}{\partial r} = \frac{\partial F}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial F}{\partial y}\cdot\frac{\partial y}{\partial r}$$

$$= (2x+3y)\cos\theta + (3x+2y)\sin\theta$$